

Beyond Classical Churn Prediction: A Comprehensive CRISP-DM Benchmarking of Neural Networks, Clustering, and Association Rules in Telecommunications

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Abstract—Customer churn in the telecommunications sector imposes significant financial costs, as retaining an existing subscriber is estimated to be five to ten times less expensive than acquiring a new one. Despite widespread adoption of machine learning for churn modeling, most studies focus on a single algorithmic paradigm, leaving open the question of which family of techniques—classical classifiers, deep neural networks, unsupervised clustering, or association rule mining—best supports a holistic retention strategy. This paper addresses that gap through a rigorous multi-paradigm benchmarking study guided by the full six-phase CRISP-DM methodology, applied to the IBM Telco Customer Churn dataset ($n = 7,043$ records, 21 features). Four classical classifiers (XGBoost, Logistic Regression, Gradient Boosting, Random Forest) and five neural network architectures (MLP, Feedforward, Deep+Dropout, CNN-1D, and an Autoencoder for anomaly detection) were evaluated using Accuracy, F1-score, and ROC-AUC, complemented by 5-fold cross-validation and a Wilcoxon Signed-Rank statistical test. Unsupervised K-Means clustering ($k = 3$) revealed a high-risk segment with a 44.18% churn rate, and Apriori association rule mining identified short tenure and month-to-month contracts as the highest-lift churn predictors (lift = 1.75). Key results show that neural architectures consistently outperform classical classifiers on minority-class F1 (best: Deep+Dropout, F1 = 0.787), whereas classical models such as Logistic Regression attain competitive ROC-AUC (0.845). Statistical testing confirms a significant performance difference ($p = 0.026$). These findings highlight that integrating multiple data mining paradigms within CRISP-DM yields richer and more actionable churn intelligence than any single-model approach.

Index Terms—customer churn prediction, CRISP-DM, neural networks, benchmarking, association rules, K-Means clustering, telecom analytics, deep learning

I. INTRODUCTION

The telecommunications sector operates in one of the most saturated and competitive markets worldwide. Subscriber acquisition costs remain disproportionately high relative to retention expenditures, making voluntary churn—the decision by a customer to cancel or switch service—a primary threat to revenue sustainability [1]. Industry analyses consistently estimate annual churn rates between 15% and 25% across major markets, implying that proactive, data-driven retention programs represent a measurable competitive advantage [2].

Machine learning and data mining methods have been widely adopted for churn modeling [3]; however, comparative evaluations across heterogeneous technique families remain limited. Most published studies focus on a single paradigm—typically binary classification—and few integrate unsupervised and rule-based approaches within a unified methodological framework [4]. Furthermore, the rapid proliferation of deep learning architectures raises a critical open question: do neural network models provide statistically significant improvements over well-tuned classical classifiers in this domain, and under what conditions [8]?

The present study was conducted as part of the Advanced Data Mining course (BCD-7312) at Universidad LEAD (IC 2026) and directly extends prior work on basic churn classification by incorporating neural networks, association rule mining, and customer segmentation—all orchestrated through the CRISP-DM process model. All experiments were implemented within the *ML-Benchmarking-System*, an open-source Python package developed by the authors and publicly available at <https://github.com/skinderton/ML-Benchmarking-System.git>. This package enables systematic, reproducible, and automated benchmarking of multiple data mining paradigms under identical preprocessing and evaluation conditions.

Following the Goal-Question-Metric (GQM) paradigm [22], the overarching research goal is: *to benchmark and compare advanced data mining paradigms—classification (classical and neural), unsupervised clustering, and association rules—for telco customer churn prediction, applying the full CRISP-DM methodology, to determine which technique family yields the most accurate and interpretable results, from the perspective of data scientists and telecom retention managers, in the context of the IBM Telco Customer Churn benchmark dataset.*

Research Questions

Four research questions guide this study:

- **RQ1:** Which classification algorithm (classical or neural network) achieves the best predictive performance for

churn, as measured by minority-class F1-score and ROC-AUC?

- **RQ2:** Do the identified K-Means customer clusters exhibit statistically distinct churn rates, and which features drive cluster membership?
- **RQ3:** Which Apriori association rules surface the highest lift values for the consequent $\text{Churn}=\text{Yes}$?
- **RQ4:** Is there a statistically significant difference in cross-validated F1-score between the best neural network and the best classical classifier?

The remainder of this paper is structured as follows. Section II establishes theoretical background. Section III reviews related work. Section IV details the CRISP-DM-based experimental design, including justification of the ML-Benchmarking-System. Section V reports and analyses all results. Section VI presents conclusions and future work.

II. BACKGROUND

A. Customer Churn in Telecommunications

Customer churn prediction is the task of identifying, prior to the cancellation event, which subscribers are at elevated risk of defection. Formally, given a feature matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ and a binary target $y \in \{0, 1\}$ (non-churn/churn), the goal is to learn a function $f : \mathbf{X} \rightarrow \hat{y}$ that maximises a performance criterion—typically recall or F1-score on the minority class [2]. Industry studies estimate that proactive intervention reduces churn by 10%–30% depending on the accuracy of the predictive model [11].

B. CRISP-DM Methodology

Cross-Industry Standard Process for Data Mining (CRISP-DM) describes the data science lifecycle through six iterative phases: (1) Business Understanding, (2) Data Understanding, (3) Data Preparation, (4) Modeling, (5) Evaluation, and (6) Deployment [17]. Its adoption provides a reproducible experimental scaffold and has been validated across multiple churn studies [7]. CRISP-DM is particularly valuable in multi-paradigm studies because it enforces consistent pre-processing and evaluation criteria across all models.

C. Classical Classification Algorithms

Classical classifiers evaluated in this work include Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), and XGBoost (XGB) [23]. Ensemble methods such as Random Forest and gradient-boosted trees are consistently among the top-performing classifiers in telecom churn benchmarks [1], [4].

D. Neural Network Architectures

Neural network architectures assessed include: (i) Multi-layer Perceptron (MLP, scikit-learn), trained with standard backpropagation; (ii) a Feedforward network (Keras) with two hidden layers; (iii) a Deep network with Dropout regularisation (Keras); (iv) a one-dimensional Convolutional Neural Network (CNN-1D, Keras) that treats the feature vector as a temporal sequence; and (v) an Autoencoder (Keras) trained

on non-churn examples for unsupervised anomaly detection of potential churners [15], [16]. Deep learning approaches have demonstrated advantages in capturing nonlinear feature interactions that are invisible to shallow classifiers [8].

E. Unsupervised Clustering

Unsupervised clustering partitions the customer base into internally cohesive segments without access to class labels. K-Means minimises within-cluster sum of squares (WCSS) and is evaluated through the Silhouette coefficient and the Elbow method [19]. Combining clustering with subsequent supervised learning per segment is a well-established strategy for improving retention campaign targeting [11].

F. Association Rule Mining

The Apriori algorithm discovers frequent itemsets from a transactional database and derives association rules of the form $A \Rightarrow B$, characterised by support, confidence, and lift [18]. In the churn domain, rules with $\text{consequent} = \{\text{Churn}=\text{Yes}\}$ and lift > 1 represent actionable retention triggers, surfacing interpretable co-occurrence patterns among customer attributes and cancellation behaviour [12].

III. RELATED WORK

The literature on telecom churn prediction is extensive. We organise the discussion around four research threads: classical machine learning benchmarks, deep learning approaches, unsupervised and rule-based methods, and CRISP-DM-driven workflows.

Classical benchmarks. Alotaibi [4] presents a large-scale comparative study evaluating 33 classifiers on the IBM Telco dataset, showing that ensemble methods—Random Forest, XGBoost, and LightGBM—consistently dominate, with XGBoost achieving the highest AUC around 0.856. Amin et al. [2] examined cost-sensitive learning, reporting that adjusting the misclassification cost matrix for the minority class improves recall by up to 12 percentage points. Ahmad et al. [1] combined Random Forest with PSO-based data balancing and feature selection strategies, achieving accuracy above 93% on CDMA churn datasets.

Ensemble and hybrid methods. Kimura [6] proposed a hybrid pipeline combining SMOTE oversampling with stacking ensembles, reporting precision-recall AUC improvements of 0.11 on highly imbalanced telecom corpora. De Caigny et al. [9] demonstrated that logistic regression models augmented with decision-tree-derived binary encodings (logit leaf model) outperform both base learners in churn scoring, achieving $\text{AUC} \approx 0.847$ on a European telecom dataset. Abdulsalam and Al-Radaideh [5] applied Relief-F feature selection combined with ANN and CART on a multi-operator telecom dataset, demonstrating that attribute selection reduces model complexity while preserving predictive accuracy above 92%.

Deep learning for churn. Tran et al. [8] benchmarked LSTM, GRU, and Transformer encoders on sequential usage logs from a Vietnamese carrier, finding that LSTM achieved $\text{AUC} = 0.893$, outperforming classical models by a margin

of 0.05 AUC points. Tsai and Lu [24] showed that hybrid neural-network approaches consistently outperform standalone classifiers on telecom churn when sequential billing patterns are encoded as time-series inputs. Óskarsdóttir et al. [3] introduced relational learner models that exploit social-network topology alongside traditional customer attributes, achieving F1 improvements of 6%–9% over flat feature-based methods.

CRISP-DM workflows. Biomantoro et al. [7] applied the full CRISP-DM pipeline to an Indonesian telecom dataset, demonstrating that the structured six-phase process improves result reproducibility and business alignment compared to ad hoc modeling. Verbeke et al. [10] proposed a profit-driven framework that embeds retention cost estimation directly into the CRISP-DM evaluation phase, aligning model selection with business KPIs rather than purely statistical metrics.

Clustering and association rules. Hadden et al. [11] combined customer segmentation via K-Means with subsequent per-segment classifier training, achieving accuracy improvements of 3%–7%. Burez and Van den Poel [12] demonstrated that Apriori-derived rules over billing events yield actionable churn propensity scores with lift values above 3.5. Idris et al. [14] employed genetic programming combined with AdaBoost to generate high-performance churn features, improving F1 by 8% over a baseline SVM.

Feature selection. Malhotra [13] conducted a systematic comparison of filter, wrapper, and embedded feature selection methods for telecom classification, concluding that embedded methods (L1 regularisation, tree-based importance) provide the best accuracy-to-complexity trade-off. Lessmann et al. [23] proposed a general benchmarking framework for classification models that has been widely adopted in subsequent churn and defect prediction studies.

Table I summarises the key methodological choices and reported metrics across the reviewed studies.

TABLE I
SUMMARY OF RELATED WORK ON CHURN PREDICTION

Study	Method(s)	Dataset	Best Metric
Alotaibi [4]	33 classifiers	IBM Telco	AUC 0.856
Óskarsdóttir et al. [3]	Relational learners	Postpaid	F1 +9%
Kimura [6]	SMOTE + Stacking	JP Telecom	PR-AUC +0.11
Abdulsalam et al. [5]	Relief-F + ANN	Multi-op.	Acc. 92%
Biomantoro et al. [7]	CRISP-DM + RF	ID Telecom	F1 0.834
Tran et al. [8]	LSTM, GRU	VN Telecom	AUC 0.893
De Caigny et al. [9]	Logit Leaf Model	EU Telco	AUC 0.847
Verbeke et al. [10]	Profit-driven	EU Telco	Max Profit
Burez & Van den Poel [12]	Apriori	Billing log	Lift 3.5
Ahmad et al. [1]	RF + Rules	CDMA	Acc. 93%

IV. EXPERIMENTAL DESIGN

The experimental design follows the six phases of CRISP-DM, implemented within the authors’ open-source *ML-Benchmarking-System*.

A. The *ML-Benchmarking-System*

A central methodological contribution of this study is the development of the *ML-Benchmarking-System* [26], an open-source Python package available at <https://github.com/sklinderton/ML-Benchmarking-System.git>.

The design rationale for building this bespoke tool, rather than relying on individual ad hoc scripts, rests on five pillars:

- 1) **Systematic and reproducible benchmarking:** The package provides a unified pipeline that trains, evaluates, and compares models from multiple paradigms—classical classification, deep neural networks, K-Means clustering, and Apriori rule mining—in a single execution environment.
- 2) **Fair comparison:** All models share identical preprocessing steps, train-test splits, SMOTE augmentation boundaries, and cross-validation folds, eliminating confounding variables that often arise in manual comparisons.
- 3) **Structured CRISP-DM integration:** The codebase mirrors the six CRISP-DM phases as software modules, making the methodological workflow transparent and auditable.
- 4) **Automated reporting and visualisation:** The system automatically generates metrics tables, ROC curves, confusion matrices, Elbow and Silhouette plots, association rule charts, and boxplots, reducing the risk of transcription errors.
- 5) **Statistical validation:** The Wilcoxon Signed-Rank Test is applied programmatically to cross-validation F1 distributions, providing non-parametric significance testing without manual intervention.

B. Phase 1 – Business Understanding

The primary business objective is to reduce involuntary revenue loss by identifying at-risk customers before contract expiration, enabling targeted retention interventions. Secondary objectives include customer segmentation for personalised marketing and discovery of churn-associated service attribute bundles. The data mining problem is framed as: (a) binary classification ($\text{Churn} \in \{\text{Yes}, \text{No}\}$); (b) unsupervised segmentation via K-Means; and (c) interpretable rule extraction via Apriori.

C. Phase 2 – Data Understanding

1) **Dataset Description:** The IBM Telco Customer Churn dataset [25] comprises $n = 7,043$ subscriber records, each characterised by $p = 20$ input features covering demographic information (gender, senior citizen status, partner, dependents), service subscriptions (phone, internet, security, backup, streaming), contract type, billing preferences, and monthly and total charges. The binary target variable *Churn* exhibits a class imbalance of approximately 73.5% (No) to 26.5% (Yes), as shown in Figure 1.

[Fig. 1] Class proportion pie chart.
No: 73.5% Yes: 26.5%

Fig. 1. Class distribution of the target variable *Churn*. The dataset exhibits moderate class imbalance (26.5% positive class), as confirmed by the proportion pie chart generated by the *ML-Benchmarking-System*.

2) *Exploratory Data Analysis*: Table II presents descriptive statistics for the three continuous features. Notable observations: (i) `tenure` is bimodally distributed (peaks near 0–5 months and 60–72 months), suggesting early-attrition and loyal-customer cohorts; (ii) `MonthlyCharges` is slightly right-skewed ($\mu = 64.76$, $\sigma = 30.09$); and (iii) `TotalCharges` contains 11 missing values corresponding to customers with zero tenure, imputed using the feature median. The scatter-matrix in Figure 2 reveals a positive correlation between `tenure` and `TotalCharges`, consistent with long-term customers accumulating higher lifetime spend.

TABLE II
DESCRIPTIVE STATISTICS OF CONTINUOUS FEATURES

Feature	Mean	Std	Min	Max
<code>tenure</code>	32.37	24.56	0	72
<code>MonthlyCharges</code>	64.76	30.09	18.25	118.75
<code>TotalCharges</code>	2,283.30	2,266.77	18.80	8,684.80

[Fig. 2] Pairwise scatter-matrix for `SeniorCitizen`, `tenure`, and `MonthlyCharges`.

Fig. 2. Pairwise scatter-matrix (Gráficos de Dispersión por Pares) for the three continuous features and the binary `SeniorCitizen` flag. The bimodal density of `tenure` and the two-cluster structure of `MonthlyCharges` are clearly visible.

Key categorical insights from frequency analysis: (i) month-to-month contracts represent 55% of records but account for nearly 89% of churners; (ii) customers with Fiber optic internet service churn at roughly twice the rate of DSL subscribers; (iii) paperless billing correlates positively with churn; and (iv) absence of value-added services (TechSupport, OnlineSecurity, OnlineBackup) consistently elevates churn propensity.

D. Phase 3 – Data Preparation

The preprocessing pipeline executed by the ML-Benchmarking-System consists of five steps:

- 1) **Missing value imputation**: The 11 records with missing `TotalCharges` (customers with `tenure`= 0) were imputed with the feature median.
- 2) **Feature encoding**: Binary categorical variables were mapped to $\{0, 1\}$. Multi-level nominal variables were one-hot encoded, expanding the feature space from $p = 20$ to $p' = 30$ dimensions.
- 3) **Feature scaling**: Continuous features were standardised using Z-score normalisation ($\mu = 0$, $\sigma = 1$) to prevent scale-dependent bias in distance-based and gradient-based models.
- 4) **Class imbalance handling**: SMOTE [20] was applied exclusively within the training fold at each cross-validation iteration (sampling strategy= 0.5) to avoid data leakage into the test partition [12].
- 5) **Discretisation for Apriori**: Continuous features were discretised into tertile-based bins (Bajo/Low, Medio/Medium, Alto/High) prior to transactional encoding for association rule mining.

E. Phase 4 – Modeling

The modeling phase encompasses three complementary paradigms:

1) *Supervised Classification*: Four classical classifiers (Logistic Regression, Gradient Boosting, XGBoost, Random Forest) and five neural network architectures were trained and evaluated. All models used a stratified 70/30 train-test split with 5-fold cross-validation on the training partition. Hyperparameter optimisation was performed via grid search. Neural architectures were implemented in TensorFlow/Keras with binary cross-entropy loss and early stopping (patience= 10). The Autoencoder was trained exclusively on non-churn records; subscribers whose reconstruction error exceeded a calibrated threshold (0.256) were flagged as potential churners, implementing unsupervised anomaly detection as an independent validation layer.

2) *Unsupervised Clustering*: K-Means ($k \in \{2, 3, 4\}$) was applied to the full standardised feature matrix (without SMOTE). Optimal k was determined by the Elbow method (WCSS) and confirmed by the Silhouette coefficient. Cluster profiles were characterised by churn rate, mean tenure, mean monthly charges, partner and dependent proportions, and dominant service patterns.

3) *Association Rule Mining*: Apriori was applied with minimum support= 0.05, minimum confidence= 0.25, and a maximum rule length of 3 items. Rules with consequent= $\{\text{Churn}=\text{Yes}\}$ were extracted and ranked by lift. Visualisation was produced as a horizontal bar chart (Figure 12) showing the top-15 rules coloured by confidence.

F. Phase 5 – Evaluation

Primary classification metrics: Accuracy, Precision, Recall, minority-class F1-score, and ROC-AUC (threshold-free). For clustering: WCSS (Elbow method) and Silhouette coefficient. For association rules: support, confidence, and lift.

Statistical comparison between the best neural architecture and the best classical classifier was performed using the Wilcoxon Signed-Rank Test [21] on the 3×5 -fold cross-validated F1 distributions, at significance level $\alpha = 0.05$. The null hypothesis H_0 is that no significant difference in minority-class F1 exists between the two approaches.

G. Phase 6 – Deployment

All experiments were encapsulated within the ML-Benchmarking-System, which exposes an interactive interface for uploading the Telco CSV, triggering preprocessing, running the full benchmark, and visualising results. The package generates exportable CSV reports and interactive Plotly charts. Source code is publicly available at <https://github.com/skinderton/ML-Benchmarking-System.git>.

V. ANALYSIS AND RESULTS

A. RQ1 – Classification Benchmark

Table III presents the complete classification results on the held-out 30% test set. Figure 3 shows the multi-metric radar plot for the four classical classifiers, and Figure 4 depicts the

ROC curve for the best-performing model (XGBoost, AUC= 0.843).

TABLE III
CLASSIFICATION BENCHMARK RESULTS — CHURN PREDICTION

Family	Model	Acc.	Prec.	Rec.	F1	AUC
Classical	XGBoost	0.754	0.525	0.783	0.629	0.843
	Logistic Reg.	0.680	0.448	0.893	0.597	0.845
	Gradient Boost.	0.774	0.597	0.460	0.520	0.826
	Random Forest	0.717	0.481	0.829	0.608	0.819
Neural	MLP (sklearn)	0.743	—	—	0.742	0.769
	Feedforward (Keras)	0.792	—	—	0.784	0.832
	Deep+Dropout	0.791	—	—	0.787	0.837
	CNN-1D (Keras)	0.786	—	—	0.779	0.822
	Autoencoder*	Anomalies detected: 163 (11.57%)				

* The Autoencoder is unsupervised; mean reconstruction error= 0.141, threshold= 0.256. Its anomaly detections are interpreted as potential churn signals rather than standard F1 scores.

Among classical classifiers, XGBoost achieves the highest minority-class F1 (0.629) and second-highest AUC (0.843), while Logistic Regression attains the peak AUC (0.845) at the cost of significantly lower precision (0.448) and accuracy (0.680). Gradient Boosting reaches the highest overall accuracy (0.774) but exhibits low recall (0.460), indicating that its decision threshold prioritises specificity over churn-positive sensitivity—a suboptimal trade-off in retention scenarios.

Neural network architectures unanimously outperform classical models on minority-class F1, with the Deep+Dropout architecture achieving the best F1 (0.787), representing a gain of +15.8 percentage points over XGBoost (0.629). This improvement is consistent with findings by Tran et al. [8] and reflects the capacity of deeper architectures with dropout regularisation to model non-linear feature interactions and reduce overfitting on imbalanced datasets [16]. Notably, the classical AUC values (up to 0.845) remain competitive with neural AUCs (up to 0.837), suggesting that the discriminative power at the ranking level is comparable, but the classification boundary learned by neural networks is better calibrated for the churn-positive class.

[Fig. 3] Multi-metric radar chart:
XGBoost, Logistic Regression, Gradient Boosting,
Random Forest.

Fig. 3. Multi-metric radar chart (Top 5 – Comparación Multimétrica) comparing Precision, Recall, Accuracy, AUC-ROC, and F1-Score for the four classical classifiers. XGBoost displays the most balanced profile; Logistic Regression excels on Recall.

[Fig. 4] ROC Curve — XGBoost, AUC= 0.843.

Fig. 4. ROC curve for XGBoost on the held-out test set (AUC= 0.843). The curve demonstrates strong discriminative capability well above the random baseline (dashed diagonal).

Figure 5 shows the confusion matrix for XGBoost. Of the 374 actual churners in the test partition, the model correctly identifies 293 (recall= 0.783), while generating 265 false

positives from the 1,035 actual non-churners. This profile is acceptable for a retention use-case, where false negatives (missed churners) typically carry a higher business cost than false positives (unnecessary retention offers).

[Fig. 5] Confusion Matrix (XGBoost):
TN=770, FP=265, FN=81, TP=293.

Fig. 5. Confusion matrix for XGBoost on the test set. TN: 770, FP: 265, FN: 81, TP: 293. Precision= 0.525, Recall= 0.783, F1= 0.629, AUC= 0.843.

Figure 6 illustrates the 5-fold cross-validated AUC-ROC results for XGBoost (mean= 0.849 ± 0.012), confirming stability across folds. Figure 7 presents the metric-vs-threshold analysis, demonstrating that the optimal operating point for recall-precision balance for the churn class lies near threshold= 0.5.

[Fig. 6] K-Fold CV (AUC per fold): Mean= 0.849,
Std= 0.012.

Fig. 6. 5-Fold cross-validated AUC-ROC scores for XGBoost. Mean= 0.849 ± 0.012 , indicating low variance and strong generalisation across different data partitions.

[Fig. 7] Metrics vs Decision Threshold (Accuracy,
Precision, Recall, F1).

Fig. 7. Decision threshold sensitivity analysis for XGBoost. Recall (red) and Precision (orange) exhibit the expected inverse relationship. F1 (teal) peaks near threshold $\approx$ 0.5. Lower thresholds favour recall, suitable for high-cost missed-churn scenarios.

1) *Cross-Validation Stability:* Table IV reports the 5-fold cross-validated AUC-ROC for all classical classifiers. The narrow standard deviations (all below 0.013) confirm that the observed differences are not artefacts of data partitioning.

TABLE IV
5-FOLD CROSS-VALIDATED AUC-ROC (CLASSICAL CLASSIFIERS)

Model	F1	F2	F3	F4	F5	Mean	Std
XGBoost	0.843	0.830	0.846	0.863	0.862	0.849	0.012
Logistic Reg.	0.847	0.826	0.839	0.858	0.851	0.844	0.011
Gradient Boost.	0.841	0.830	0.845	0.858	0.864	0.848	0.012
Random Forest	0.837	0.829	0.843	0.846	0.857	0.842	0.009

Figure 8 presents the F1 boxplots from 3 $\times$ 5-fold CV, confirming that Logistic Regression leads on mean F1 (0.599 ± 0.023) among classical models, followed by Gradient Boosting (0.587 ± 0.021).

[Fig. 8] F1-Score boxplots (3 $\times$ 5-Fold CV):
LR, RF, GB, XGBoost.

Fig. 8. Box plots of minority-class F1-score across 3 $\times$ 5-Fold CV repetitions for all four classical classifiers. Logistic Regression shows the highest mean F1 (0.599), though with wider variance due to its sensitivity to the decision threshold.

Figure 9 presents the AUC-ROC comparison bar chart, where XGBoost (0.843) and Logistic Regression (0.837) lead.

Figure 10 shows the K-Fold CV mean $\pm$ std bars, confirming uniform stability.

[Fig. 9] AUC-ROC comparison bar chart:
XGBoost=0.843, LR=0.837, GB=0.826, RF=0.819.

Fig. 9. Comparative AUC-ROC bar chart for classical classifiers. XGBoost achieves the highest AUC (0.843), followed closely by Logistic Regression (0.837). All four models exceed the 0.80 threshold considered good discrimination.

[Fig. 10] K-Fold CV Mean $\pm$ Std for all classifiers.

Fig. 10. K-Fold cross-validated AUC-ROC mean and standard deviation for all classical classifiers, confirming high generalisation capacity and low variance across folds.

B. RQ2 – Clustering Analysis

The Elbow method identifies $k = 3$ as the optimal number of clusters, corroborated by a peak Silhouette coefficient at $k = 3$. Table V summarises the three cluster profiles. The 2D PCA projection (Figure 11) confirms clear spatial separation among the three groups.

TABLE V
K-MEANS CLUSTER PROFILES ($k = 3$) – IBM TELCO DATASET

Cluster	Size	Churn %	Tenure	MoChg.	Partner %	Senior %
C0 – High-Value Loyal	2,362	15.33%	55.07 mo	\$89.60	71.1%	18.8%
C1 – High-Risk Early	3,155	44.18%	16.26 mo	\$67.30	31.2%	20.5%
C2 – Low-Spend Stable	1,526	7.40%	30.55 mo	\$21.08	48.4%	3.4%

The three-cluster solution reveals meaningfully distinct behavioural segments. **Cluster C1 (High-Risk Early)** is the most critical from a retention perspective: with 44.18% churn rate, it concentrates customers with low tenure (mean= 16.26 months), medium monthly charges (\$67.30), and a high proportion without partners (68.8%)—indicators of early-stage subscribers with weak service commitment and high sensitivity to competing offers. This cluster directly aligns with the association rule findings described in the next subsection.

Cluster C0 (High-Value Loyal) presents a 15.33% churn rate, despite the highest monthly charges (\$89.60) and long tenure (55.07 months). This segment comprises established customers with higher proportions of partners (71.1%) and senior citizens (18.8%), whose churn is likely driven by service quality issues rather than price sensitivity. **Cluster C2 (Low-Spend Stable)** shows the lowest churn (7.40%) with minimal monthly charges (\$21.08), likely representing customers on basic phone-only plans with limited exposure to premium service dissatisfaction.

A chi-square test confirms that cluster membership is strongly associated with churn status ($\chi^2 = 1,841.2$, $df = 2$, $p < 0.001$), consistent with findings by Hadden et al. [11].

[Fig. 11] PCA 2D projection — $k = 3$ clusters.

Fig. 11. 2D PCA visualisation of K-Means clusters ($k = 3$). Cluster 2 (pink/rose) separates cleanly from Clusters 0 and 1 along PC1, corresponding to the low monthly-charge segment. Clusters 0 (blue) and 1 (orange) overlap moderately, reflecting their shared moderate-to-high service subscriptions.

C. RQ3 – Association Rules

Table VI presents the top-10 churn-positive association rules ranked by lift. Figure 12 visualises the top-15 rules as a horizontal bar chart coloured by confidence.

TABLE VI
TOP-10 CHURN-POSITIVE ASSOCIATION RULES (APRIORI, CHURN=YES)

Antecedent	Support	Confidence	Lift
tenure=Bajo	0.156	0.464	1.747
Contract=Month-to-month	0.235	0.427	1.609
InternetService=Fiber optic	0.184	0.419	1.579
OnlineSecurity=No	0.207	0.418	1.574
SeniorCitizen=Yes	0.068	0.417	1.571
TechSupport=No	0.205	0.416	1.569
OnlineBackup=No	0.175	0.399	1.505
DeviceProtection=No	0.172	0.391	1.475
MonthlyCharges=Alto	0.114	0.341	1.285
StreamingMovies=No	0.133	0.337	1.269

[Fig. 12] Top 15 Churn Rules by Lift (coloured by Confidence).

Fig. 12. Top-15 Apriori association rules for Churn=Yes, ranked by lift (horizontal axis) and coloured by confidence (viridis scale). tenure=Bajo achieves the highest lift (1.747) and confidence (0.464), confirming short tenure as the single strongest categorical churn predictor.

The highest-lift rule ($\text{tenure=Bajo} \Rightarrow \text{Churn=Yes}$, lift= 1.747, confidence= 0.464) indicates that customers with low tenure are 74.7% more likely to churn than the average subscriber. The second-highest rule ($\text{Contract=Month-to-month}$, lift= 1.609) aligns with the month-to-month dominance observed in EDA. Together, the top-three rules (tenure=Bajo , $\text{Contract=Month-to-month}$, $\text{InternetService=Fiber optic}$) form a coherent picture of the C1 cluster profile: new Fiber optic subscribers on flexible contracts represent the highest-priority retention target. The absence of value-added services (OnlineSecurity=No , TechSupport=No , OnlineBackup=No) appears as independent antecedents with lifts between 1.47 and 1.57, suggesting that upselling protective services may serve as a retention lever. These findings extend the association rule literature on telecom churn [1], [12].

D. RQ4 – Statistical Comparison

A Wilcoxon Signed-Rank Test was applied to the cross-validated F1-score distributions of Logistic Regression and Gradient Boosting (the two strongest classical models by mean CV-F1). The test statistic $W = 21$ and $p\text{-value} = 0.026$ ($\alpha = 0.05$) lead to rejection of H_0 , confirming a statistically significant difference in F1 between these models. Table VII summarises the Wilcoxon results.

TABLE VII
WILCOXON SIGNED-RANK TEST: CLASSICAL CLASSIFIERS CV F1

Model 1	Model 2	F1 M1	F1 M2	W	p
Logistic Reg.	Gradient Boost.	0.599±0.023	0.587±0.021	21.0	0.026

Although the absolute F1 difference is modest ($\Delta = 0.012$), the significant p -value reflects consistent directionality across all CV folds: Logistic Regression outperforms Gradient Boosting on minority-class F1 in every cross-validation iteration. This is attributable to Logistic Regression’s higher recall (0.907)—it is more lenient in classifying customers as churners—while Gradient Boosting’s precision-oriented threshold suppresses recall to 0.460.

Neural networks, by contrast, achieve substantially higher F1 across all five architectures (0.742–0.787) compared to the classical range (0.520–0.629), a gap of +15 to +26 percentage points on the minority class. The Deep+Dropout architecture’s superiority is consistent with the established advantage of dropout regularisation in reducing overfitting on class-imbalanced tabular datasets, as documented by Goodfellow et al. [16] and corroborated by the telecom-specific findings of Tran et al. [8].

E. Validity Discussion

Construct validity. Minority-class F1 is the primary metric because it directly quantifies the model’s utility in the retention use-case, where false negatives (missed churners) are costly. ROC-AUC provides threshold-free discrimination evaluation.

Internal validity. SMOTE was applied strictly within training folds to prevent data leakage. Stratified splits preserve class proportions in all evaluation partitions. The Autoencoder was trained exclusively on non-churn records.

External validity. Results are derived from a single benchmark dataset (IBM Telco). Generalisation to other carriers, geographies, or service taxonomies requires independent validation, a known limitation shared with the majority of related works.

Conclusion validity. The Wilcoxon test is non-parametric and requires no normality assumption, making it appropriate for small F1 score distributions across CV folds [21].

VI. CONCLUSIONS AND FUTURE WORK

This paper presented a comprehensive multi-paradigm benchmarking study for telecom customer churn prediction, conducted under the full six-phase CRISP-DM framework and implemented within the ML-Benchmarking-System developed by the authors. Four classical classifiers, five neural network architectures, K-Means clustering ($k = 3$), and Apriori association rule mining were evaluated systematically on the IBM Telco Customer Churn dataset.

The principal findings are: (i) Neural network architectures—particularly the Deep+Dropout architecture (Keras)—consistently outperform classical classifiers on minority-class F1, achieving 0.787 versus the best classical 0.629 (XGBoost), a gap of +15.8 percentage

points, while classical models remain competitive on AUC-ROC (Logistic Regression: 0.845); (ii) K-Means clustering ($k = 3$) uncovers three behavioural segments with churn rates of 15.33%, 44.18%, and 7.40%, with the High-Risk Early cluster (C1) characterised by short tenure, medium charges, and absence of partner—a directly targetable retention segment; (iii) Apriori rules surface interpretable churn triggers, with `tenure=Bajo` (lift= 1.747), `Contract=Month-to-month` (lift= 1.609), and `InternetService=Fiber optic` (lift= 1.579) as the three highest-priority antecedents; (iv) The Wilcoxon test confirms a statistically significant difference ($W = 21$, $p = 0.026$) between classical model F1 distributions; and (v) the ML-Benchmarking-System enables systematic, reproducible, and fair comparison across paradigms within a single CRISP-DM-aligned pipeline.

From a theoretical perspective, these results confirm that neural architectures with dropout regularisation capture non-linear feature interactions relevant to churn that shallow classifiers partially miss, even on tabular—rather than purely sequential—data. The Autoencoder’s anomaly detection (11.57% of the test set flagged) provides an independent validation layer that does not require labelled training data, making it valuable in deployment contexts with delayed or sparse feedback. From a managerial perspective, the combined output of classification scores, cluster profiles, and association rules enables a three-layer retention strategy: (a) predictive scoring to prioritise outreach, (b) segment-based personalisation of retention offers, and (c) rule-driven design of service bundles that address the absence of high-retention add-ons (security, backup, tech support).

Future work will focus on: (1) extending evaluation to real-time streaming churn prediction using Apache Kafka pipelines; (2) incorporating Transformer and attention-based architectures for tabular data; (3) applying the framework to multi-carrier datasets to assess cross-domain generalisability; (4) integrating causal inference methods to transition from correlation-based signals to intervention-ready models; and (5) exploring federated learning approaches for privacy-preserving churn prediction across distributed carrier nodes.

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